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Logic circuitry package for print apparatus

Abstract

A logic circuitry package for a print apparatus includes a serial data bus interface to interface with a serial data bus, the logic circuit to respond to messages from the print apparatus over the serial data bus directed to a number of different addresses using different component keys for cryptographic authentication corresponding to each different address, the different component keys related to at least one master key of the print apparatus. The logic circuitry package further includes control logic which, in response to a session initiation request message on the serial data bus from the print apparatus, is to establish a secure communication session related to one of the different addresses, the control logic to cryptographically authenticate responses to each subsequent message of the secure communication session from the print apparatus using a component key corresponding to the address.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This Utility Patent Application is a U.S. National Stage filing under 35 U.S.C. § 371 of PCT/US2020/030860, entitled LOGIC CIRCUITRY PACKAGE FOR PRINT APPARATUS, filed Apr. 30, 2020, incorporated by reference herein.

BACKGROUND

(2) Some 2D and 3D printing systems include a number of replaceable print apparatus components, such as inkjet printhead assemblies, and print material containers (e.g. inkjet cartridges, toner cartridges, ink supplies, build material supplies, etc.), for example. In some cases, such replaceable print apparatus components include logic circuitry to communicate with logic circuitry of the print apparatus in which they are installed. Information communicated by replaceable print material containers, for example, may include an identity (such as a serial number and/or brand, for example), identifying characteristics (such as a color, color mapping, and a color map reconstruction, for example), and operating information (such as a print material level and an operating pressure).

(3) Communication between the print apparatus and associated replaceable print apparatus

components may be carried out using serial data bus protocols, such as Serial Peripheral Interface (SPI) protocol, or Inter-integrated circuit (I²C or I2C) protocol, where at least one “master” integrated circuit (IC) communicates with at least one “slave” IC via a serial data bus. In a case where a printing system utilizes I²C communications, the print apparatus may include a master IC, and each replaceable print apparatus component may include a slave IC, with each slave IC having a corresponding I²C address and communicating with the master IC over a shared I²C serial data bus. The slave ICs respond to requests (commands) from the master IC, and may include logic circuitry to carry out functions associated with the operation of the replaceable print apparatus component.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block and schematic diagram generally illustrating a logic circuitry package for a print apparatus, according to one example.

(2) FIG. 2 is a block and schematic diagram generally illustrating a print apparatus including a logic circuitry package, according to one example.

(3) FIG. 3 is a block and schematic diagram generally illustrating a logic circuitry package for a print apparatus, according to one example.

(4) FIG. 4A is a block and schematic diagram generally illustrating a flexible wiring substrate for connecting a logic circuitry package to a print apparatus, according to one example.

(5) FIG. 4B is a block and schematic diagram generally illustrating a flexible wiring substrate for connecting a logic circuitry package to a print apparatus, according to one example.

DETAILED DESCRIPTION

(6) In the following detailed description, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific examples in which the disclosure may be practiced. It is to be understood that other examples may be utilized and structural or logical changes may be made without departing from the scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense, and the scope of the present disclosure is defined by the appended claims. It is to be understood that features of the various examples described herein may be combined, in part or whole, with each other, unless specifically noted otherwise.

(7) Some 2D and 3D printing systems include a number of replaceable or consumable print apparatus components. Example 2D and 3D printing systems include two-dimensional imaging apparatus and three-dimensional additive manufacturing apparatus, such as inkjet printers, dry toner printers, liquid toner printers, and 3D powder bed inkjet printers, for example. Example replaceable print apparatus components include any number of print material containers, such as ink tanks, ink bottles, inkjet printhead cartridges, dry toner reservoirs, liquid toner reservoirs, dry toner cartridges, and build material supplies; and other components that may or may not contain print materials, such as printheads (e.g., a thermal inkjet printhead).

(8) In some cases, such replaceable print apparatus components include logic circuitry to communicate with logic circuitry of the print apparatus in which the replaceable print apparatus components are installed. In the case of a replaceable print apparatus component being a replaceable print material container, for example, information communicated by a replaceable print material container with the print apparatus may include an identity (such as a serial number and/or brand, for example), identifying characteristics (such as a color, color mapping, and a color map reconstruction, maximum print agent volume, and functionality, for example), and operating information (such as a print material level and an operating pressure, for example).

(9) Communication between the print apparatus and replaceable print apparatus components may

be carried out using serial data bus protocols, such as SPI protocol or I2C protocol, where at least one master IC communicates with at least one slave IC over a serial data bus. For example, where a printing system utilizes I2C communications, logic circuitry of the print apparatus may operate as a master IC, while logic circuitry of each replaceable print apparatus component may operate as a slave IC, with each slave IC having a corresponding I2C address and communicating with the master IC over the I2C serial data bus.

(10) Slave ICs may include logic circuitry to perform data operations associated with the replaceable print apparatus component and to respond to requests (commands) from the master IC via the I2C serial data bus. For example, the print apparatus, via the master IC, may transmit commands over the I2C serial data bus to the slave ICs to perform various operations. A request may comprise a “write” request (e.g., to write data to a memory location), or a request to perform an action, such as performing a measurement (e.g., ink level measurement), carrying out a printing task, and a data processing action, for example. In examples, a master IC (as part of a printer controller) may send a command to a slave IC (as part of a replaceable print apparatus component) to perform an operation to load data into a memory (e.g., a buffer and/or specific register of a memory), and, in response to a subsequent print apparatus command, the slave IC may transmit the data over the I2C serial data bus.

(11) In some cases, to ensure the quality of a replaceable print apparatus component, and thereby ensure proper operation of the print system, the print apparatus may perform validation processes to verify that a replaceable print apparatus component originates from an authorized source. For example, a print apparatus command (a “read” request) may request identification and/or authentication information to validate the replaceable print apparatus component, which may result in the replaceable print apparatus component loading such information into a memory buffer, and subsequently serially transmitting such information via the I2C serial data bus. In some cases, to perform such validation, communications between the master IC and each slave IC may be cryptographically authenticated. In one case, see, for example, U.S. Pat. No. 9,619,663, the slave ICs may include a number of secret keys derived from a number of secret keys of the master IC of the print apparatus, where the secret keys are used to generate cryptographically authenticated communications between the master and slave ICs.

(12) Further authentication mechanisms may also be employed, such as logic and dedicated hardware for providing responses within predetermined time windows, such as described, for example, by U.S. Pat. No. 9,561,662. Other validation processes may include integrity checks to ensure that a replaceable print apparatus component and/or the logic circuitry (e.g. slave IC) associated therewith is functioning as expected.

(13) In some cases, a replaceable print apparatus component may include more than one slave IC, such as two slave ICs, for example, where communication by a master IC with a first one of the two slave ICs may be cryptographically authenticated, while communications with the second of the two slave ICs may be non-authenticated.

(14) While logic circuitry onboard each replaceable print apparatus component is effective at providing authenticated communication between the print apparatus and the replaceable print apparatus component, providing logic circuitry on each replaceable print apparatus component is costly.

(15) According to one example, the present disclosure provides a single logic circuitry package that is connectable to secure firmware of the print apparatus and provides cryptographically authenticated responses to the secure firmware in response to messages directed to a number of different I2C addresses using different secret keys corresponding to each of the addresses. In one example, the single logic circuitry package is implemented separately from any of replaceable print apparatus components. In other examples, the single logic circuitry package may reside in one of the replaceable print apparatus components, such as part of a print material container (e.g., a black ink supply). In one example, the single logic circuitry package communicates with the secure

firmware of the print apparatus over an I2C serial bus.

(16) Employing a single logic circuitry package representing multiple I2C addresses and using multiple keys for cryptographic authentication, in accordance with the present disclosure, may reduce the costs of replaceable print apparatus components. By employing a single logic circuitry package to respond to secure communications directed to multiple addresses using multiple keys, separate logic circuitry may not need to be attached to each individual replaceable print apparatus component of a printing system.

(17) FIG. 1 is a block and schematic diagram generally illustrating a logic circuitry package **20** for a print apparatus **10**, according to one example of the present disclosure. Logic circuitry package **20** interfaces with a serial data bus **12** of print apparatus **10** via a serial data bus interface **22**, and is configured to respond to messages on serial data bus **12** directed to a number of different addresses **26** using different component keys **28** for cryptographic authentication corresponding to each of the different addresses **26**, where the different component keys **28** are related to at least one master key **14** of print apparatus **10**.

(18) In one example, logic circuitry package **20** is a single package component having a single data contact to interface with serial data bus **12** via which messages to the number of different addresses **26** are received and corresponding responses to such messages are transmitted. In one example, serial data bus interface **22** is an I2C serial data bus interface, where the different addresses **26** are I2C addresses.

(19) In examples, in response to a session request message on serial data bus **12** from print apparatus **10** to establish a secure communication session related to one of the different addresses **26**, control logic **30** of logic circuitry package **20** establishes such a secure communication session and cryptographically authenticates responses to each subsequent message of the communication session from print apparatus **10** using one of the component keys **28** corresponding to the one of the different addresses **26**. As will be described in greater detail below, in examples, for each of the different addresses **26**, in response to receiving a session request message to establish a communication session with the address, control logic **30** generates a component session key **40** and a component session key identifier **42** based on a component key **28** corresponding to the address **26**, and provides a session initiation response message including the component session key identifier **42** on serial data bus **12**.

(20) In other examples, as will be described in greater detail below, for each of the different addresses **26**, after establishment of a secure communication session via a session initiation request/session initiation response message exchange, in response to receiving an authenticated request message of the communication session directed to the address, where such authenticated request message includes a first message authentication code (MAC) based on a session key associated with master key **14** and on the component session key identifier **42** included in the session initiation response message, control logic **30** generates a second MAC related to component session key **40** and the first MAC and provides a cryptographically authenticated response message including the second MAC along with the data requested by the authenticated request message.

(21) In examples, print apparatus **10** may include a plurality of consumable or replaceable print apparatus components **16**, illustrated as replaceable print apparatus components **16a** to **16n**, where the number of different addresses **26** correspond to different ones of the replaceable print apparatus components **16**. Replaceable print apparatus components **16** may include any number of replaceable components such as print material dispensing components (e.g., printheads), and print material containers such as cartridges for storing liquid print agent (e.g., ink) for 2D and 3D printing, toner, and 3D print agent or build material, for example. As will be described in greater detail below, in accordance with examples of the present disclosure, logic circuitry package **20** maintains operating data related to each of the different addresses **26**, such as operating data associated with replaceable print apparatus components **16** where the different addresses **26**

correspond to such replaceable print apparatus components **16**. In one example, such operating data may include component keys **28**, component identification information, and operational and non-operational parameters for both the replaceable component and the print apparatus.

(22) In accordance with examples of the present disclosure, by employing a single logic circuitry package **20** to provide cryptographically authenticated communication related to a number of different addresses using different component keys for each address, it may not be necessary for various replaceable print apparatus components, such as replaceable print apparatus components **16**, for example, to include such logic circuitry (e.g., microcontrollers), thereby reducing the costs of such print apparatus components.

(23) FIG. **2** is a block and schematic diagram generally illustrating a print apparatus **10** employing a logic circuitry package **20**, in accordance with the present disclosure, to carry out cryptographically authenticated communications related to a number of different addresses **26** (see FIG. **3**, which illustrates one example of logic circuitry package **20**, an example of the operation of which is described in conjunction with FIG. **2** below). Continuing with FIG. **2**, in one example, print apparatus **10** includes a printer controller **60**, a serial data bus **80**, and a number of replaceable print apparatus components **16**, illustrated as replaceable print apparatus components **16a** to **16d**. In one example, the number of different addresses **26** comprises a number of different first default addresses (see **102** in FIG. **3**), with each of the different first default addresses corresponding to a different one of the replaceable print apparatus components **16a** to **16d**. In one example, as will be described in greater detail below, the number of different addresses **26** further includes second default addresses (see **104** in FIG. **3**) corresponding to the different first default addresses.

(24) In one example, printer controller **60** includes secure firmware **62** (e.g., a programmed microprocessor), a master IC **64** controlled by firmware **62**, and a memory **66** for storing operating information, such as an operating program and at least one master key **14**. Secure firmware **62** communicates with various components of print apparatus **10** on serial data bus **80**, including directing messages (commands) to each of the different first default addresses of replaceable print apparatus components **16a** to **16d**, over serial data bus **80** via master IC **64**. In one example, secure firmware **62** may reside within master IC **64**. In one example, memory **66** may comprise a secure microcontroller including secure memory storage.

(25) As described in greater detail below, logic circuitry package **20** is configured to respond to messages directed by printer controller **60** over serial data bus **80** to the each of the different first default addresses using different component keys **28** corresponding to each different first default address for cryptographic authentication, where the different component keys **28** are related to at least one master key **14** of print apparatus **10**.

(26) In one example, logic circuitry package **20** is disposed as part of one of the replaceable print apparatus components **16**, such as replaceable print apparatus component **16a**. In other examples, portions of logic circuitry package **20** may be disposed in multiple of the replaceable print apparatus components **16a** to **16d**. In other examples, logic circuitry package **20** may be disposed within print apparatus **10** separately from replaceable print apparatus components **16**. In other examples, a portion of logic circuitry package **20** may be disposed within at least one of the replaceable print apparatus components **16**, and another portion may be disposed separately from replaceable print apparatus components **16**. In one example, logic circuitry package **20** is a single package component having a single contact point to interface with serial data bus **80** and being configured to receive messages directed to each of the first default addresses of the replaceable print apparatus components **16** and to transmit responses to such messages via the single contact point.

(27) In one example, print apparatus **10** includes a print material dispensing apparatus **72** to dispense print material onto a print media **74** conveyed by a media transport mechanism **76**. In one example, each of the replaceable print apparatus components **16a** to **16d** comprises a print material cartridge which provides print material stored within a corresponding reservoir **90a** to **90d** to print

material dispensing apparatus 72. In one example, print material dispensing apparatus 72 is an inkjet printhead controlled by printer controller 60 to eject drops of ink in a desired pattern to form a desired print image on print media 74, with each print material cartridge 16a to 16d being an ink cartridge holding ink in a corresponding reservoir 90a to 90d, where controller 60 pressurizes ink reservoirs 90 via a pump 78 to supply ink from print material cartridges 16a to 16d to inkjet printhead 72.

(28) In examples, each print material cartridge 16a to 16d may hold a different ink color. In one case, reservoir 90a of print material cartridge 16a may hold black ink, reservoir 90b of print material cartridge 16b may hold cyan ink, reservoir 90c of print material cartridge 16c may hold magenta ink, and reservoir 90d of print material cartridge 16d may hold yellow ink.

(29) In one example, print apparatus 10 includes a number of cartridge slots 92a to 92d into which print material cartridges 16a to 16d may be selectively inserted or removed. In one example, each of the cartridge slots 92a to 92d corresponds to a particular ink color. Continuing with the above example, in one instance, cartridge slot 92a is a black ink slot to hold black print material cartridge 16a, cartridge slot 92b is a cyan ink slot to hold cyan print material cartridge 16b, cartridge slot 92c is a magenta ink slot to hold magenta print material cartridge 16c, and cartridge slot 92d is a yellow ink slot to hold yellow print material cartridge 16d.

(30) In one example, each cartridge slot 92 includes a set of contacts 94, illustrated as contacts 94a to 94d, which connect cartridge slots 92a to 92d to serial data bus 80. In one example, each print material cartridge 16a to 16d includes a set of contacts 96, illustrated as contacts 96a to 96d, which couple with the respective set of contacts 94a to 94d of the corresponding cartridge slot 92a to 92b to connect the print material cartridges 16a to 16d to serial data bus 80. In one example, logic circuitry package 20 disposed within print material cartridge 16a is connected to serial data bus 80 via contacts 94a and 96a and a communication link 98. In other example, contacts 96b to 96d may be absent from print material cartridges 16b to 16d not including logic circuitry package 20.

(31) In one example, serial data bus 80 is an I2C serial data bus, such that the corresponding first and second default address of each print material cartridge 16a to 16d is an I2C address. In the illustrated example, I2C serial data bus 80 includes a clock line SCL, a bidirectional data line SDA, a ground line, GND, and a power line Vdd, connected to a voltage source 82. Data line SDA is connected to a voltage source 84 via a pull-up resistor 86. In one example, a series of divider resistors, illustrated as divider resistors 88a to 88d, are connected to form a resistive ladder network with divider resistor 88a being connected between cartridge slot 92a and printer controller 60, divider resistors 88a and 88b being series-connected between cartridge slot 92b and printer controller 60, divider resistors 88a-88c being series-connected between cartridge slot 92c and printer controller 60, and divider resistors 88a-88d being series-connected between cartridge slot 92d and printer controller 60. In one example, as will be described in greater detail below, such a resistive ladder network may be employed by printer controller 60 to determine a cartridge slot 92a to 92d in which a print material cartridge 16a to 16d is inserted. It is noted that, in other examples, divider resistor 88a may not be employed between printer controller 60 and first cartridge slot 16a.

(32) FIG. 3 is a block and schematic diagram generally illustrating logic circuitry package 20, according to one example. Logic circuitry package 20 includes an I2C serial interface 22 to interface with I2C bus 80 (see FIG. 2), control logic 30, a non-volatile memory 100, and a volatile memory 120. In one example, non-volatile memory 100 stores the first default I2C address 102 and the second default I2C address 104 of each print material cartridge 16a to 16d (see FIG. 2). In one example, in lieu of storing first and second default I2C addresses 102 and 104, upon each power-up, logic circuitry package 20 algorithmically derives first default I2C addresses 102 and second default I2C addresses 104 from values stored in nonvolatile memory 100 (e.g., in a lookup table) which are representative of first and second default I2C addresses 102 and 104. In another example, upon each power-up, logic hardware of logic circuitry package 20 generates first and second default I2C addresses 102 and 104.

(33) In one example, for each first default I2C address **102**, nonvolatile memory **100** stores a corresponding unique ID **106** representative of an ID of the corresponding print material cartridge **16a** to **16d**. In one example, for each first default I2C address **102** representing print material cartridges **16a** to **16d**, non-volatile memory **100** stores a number of component keys **28** and a corresponding component key identifier **108** for each component key **28**, where each component key **28** is related to (e.g., derived from) a corresponding master key **14** stored in printer controller **60**. For example, in one case, each first default I2C address **102** may have **8** component keys **28**, with each of the **8** component keys **28** being related to a different one of **8** master keys **14** stored in printer controller **60**. In one example, for each first default I2C address **102**, non-volatile memory **100** further stores operating data representative of operating data of the corresponding print material cartridge **16a** to **16d**. In one example, logic circuitry package **20** stores master keys **14-1** related to master keys **14** (e.g., identical) of printer controller **60**. In one example, based on the respective unique IDs, logic circuitry package **20** calculates the component keys **28** for each first default I2C address **102** from master keys **14-1**.

(34) An example operation of logic circuitry package **20** is described below with reference to FIGS. 2 and 3. When print apparatus **10** is not performing a “print job” (i.e., not printing images to print media **74**), logic circuitry package **20**, along with print material cartridges **16a** to **16d**, are powered down, such that previously stored information in volatile memory **120** is not maintained. Upon commencement of a print job, print material cartridges **16a** to **16d**, along with logic circuitry package **20**, are powered up such that logic circuitry package **20**, and other components on serial data bus **80**, are enabled to communicate with printer controller **60**.

(35) I2C commands directed by printer controller **60** to I2C addresses of print apparatus components on serial data bus **80** are carried out in the form of I2C data packets or frames, where a data packet begins with a START condition (e.g., a fall in voltage on the SDA line while the SCL line is high), and ends with a STOP condition (e.g. a rise in voltage on the SDA line while the SCL line is high). Between the START and STOP conditions, the packet includes a series of data bytes, where a first data byte represents the I2C address of the component to which the command is being directed. Subsequent data bytes of the packet may define a command code (indicative of an action to be performed by the component), and additional command data (e.g., additional command parameters, write data, and MACs).

(36) As will be described in greater detail below, during a print job, logic circuitry package **20** monitors communications on serial data bus **80** from printer controller **60** and maintains address-specific communication sessions with printer controller **60** for each of the first and second I2C addresses which emulate expected responses and functionalities of the corresponding print material cartridges **16a** to **16d**. As will be described in greater detail below, in one example, for each of the print material cartridges **16a** to **16d**, it is noted that only one of the first and second I2C addresses is active at a given time.

(37) In examples, upon recognizing an I2C START condition of an I2C data packet on serial data bus **80**, logic circuitry package **20** receives the subsequent address byte and compares the address represented by the address byte with the first default I2C addresses **102** representing print material cartridges **16a** to **16d**. If there is no match, logic circuitry package **20** waits for the next I2C start condition.

(38) If there is a match between the I2C address represented by the address byte and one of the first default I2C addresses **102**, logic circuitry package **20** receives the subsequent command data bytes. In one example, a data byte following the address byte represents a command code, where logic circuitry package **20** includes, such as within non-volatile memory **100**, a command code look-up table for each of the first default I2C address **102** which provides instructions for control logic **30** to carryout operations to satisfy the designated command. In one example, the command code is indicative of whether a command from printer controller **60** is authenticated. In other examples, control logic **30** performs the operations and returns a response message including data responsive

to the designate command to printer controller **60** via serial data bus **80**.

(39) In examples, each first default I2C address corresponding to print material cartridges **16a** to **16d** has a number of corresponding I2C commands to which logic circuitry package **20** responds. In one case, the requested I2C command by printer controller **60** may be a “read” command to read the unique ID **106** corresponding to one of the first default I2C addresses **102** designate by the read command. In one case, upon initiation of a print job, printer controller **60** may issue a series of read commands directed to each of the first default I2C addresses **102** to read the corresponding unique ID from a designated memory location, where logic circuitry package **20** carries out each read request and provides a separate response to each read command including the requested unique ID. In one example, based on the respective unique ID, master IC **64** of printer controller **60** calculates the component keys **28** for each first default I2C address **102** from master keys **14**, as indicated by component keys **28-1** in memory **66** (see FIG. 2).

(40) In one case, during a print job, to verify an authenticity of print material cartridges **16a** to **16d** represented by corresponding first default I2C addresses **102**, printer controller **60** may issue a “Start Session” command to initiate a secure communication session with each of the first default I2C addresses **102**. As an illustrative example, printer controller **60** may issue a “Start Session” command directed to the first default I2C address **102** for yellow print material cartridge **16d**. After the START condition on serial data bus **80**, and after recognizing that that I2C address represented by the subsequent address byte matches the first default I2C address **102** corresponding to yellow print material cartridge **16d**, logic circuitry package receives the subsequent command bytes and determines, such as via the command code and lookup table, that the command is a “Start Session” command.

(41) In one example, the “Start Session” command data includes data designating a particular component key **28** of a number of component keys corresponding to the first default I2C address **102** representing yellow print material cartridge **16d**, and a host value (which may be a random value generated by master IC **64**). In response, according to one example, logic circuitry package **20** generates a component value corresponding to the first default I2C address **102** representing yellow print material cartridge **16d** (such as a counter value which is incremented for each “Start Session” command). In one example, logic circuitry package **20** constructs a component session key identifier **42** based on the component key identifier **108** of the designated component key **28**, the host value, and the component value. In one example, logic circuitry package **20** then derives a component session key **40** from the designated component key **28** and the component session key identifier **42**. In one example, logic circuitry package **20** stores component session key **40** in volatile memory **100**, and returns component session key identifier **42** in a response message to printer controller **60**.

(42) Upon receiving the response message, master IC **64** derives a master session key **68** from the component session key identifier **42** and the component key **28-1** as designated by the “Start Session” command, where master session key **68** should match the component session key **40**. Thereafter, printer controller **60** sends an “authenticated” command directed to the first default I2C address **102** for yellow print material cartridge **16d**, where the authenticated command includes a first MAC computed using the corresponding master session key **68**. As an illustrated example, the authenticated command may be a “read” command to read data from a specified memory location corresponding to the first default I2C address **102** for yellow print material cartridge **16d**, where the read command includes the designated I2C address, a command code, command parameters (e.g., a designated memory location and how many bytes to read), and a first MAC, where the first MAC is computed by master IC **64** using the corresponding Master Session Key **68**, the command code, and the command parameters.

(43) In response, logic circuitry package **20** reads the data from the designated memory location, and provides a response message including the read data from the designated location and a second MAC, where the second command code is calculated by controller **30** using the read data, the

Component Session Key **40**, and the first MAC. Upon receiving the response message from logic circuitry package **20**, printer controller **60** computes a third MAC using the read data in the response message, the corresponding Master Session Key **68**, and the first MAC. Printer controller **60** then compares the third MAC with the second MAC to determine the authenticity of the responses to commands directed to the first default I2C address **102** for yellow print material cartridge **16d**. If the third MAC matches the second MAC, printer controller **60** deems the responses to commands directed to the first I2C address **102** for yellow print material cartridge **16d** to be authenticated and that the read data provided by the response message can be trusted.

Conversely, if there is a mismatch, printer controller **60** deems the responses to commands directed to the first default I2C address **102** for yellow print material cartridge **16d** to be invalid.

(44) In examples, after initiation of the secure communication session via the “Start Session” command, the above authentication process is performed for each command message directed to the first I2C address **102** for yellow print material cartridge **16d** for a remainder of the secure communication system. In this regard, component session key **40** is maintained within volatile memory **120**, with the first, second, and third MACs being generated for each command-response message pair related to the first default I2C address **102** for yellow print material cartridge **16d**.

(45) In examples, an authentication process as described above is carried out with printer controller **60** by logic circuitry **60** to establish secure, cryptographically authenticated communication sessions with each of the first default I2C addresses **102**, with the derived component session key **40** for each of the first default I2C addresses **102** being stored in volatile memory **120** for a duration of the print job.

(46) In examples, after a secure communication session with a first default I2C address **102** has been established, printer controller **60** may direct an “activate address” command to the first default I2C address **102** which includes a designated time duration. Upon receipt of the “activate address” command, logic circuitry package **20** is to not respond to commands directed to the corresponding first default I2C address **102** for the designated time duration, but to instead provide responses to commands directed to the corresponding second default I2C address **104**, where such communication is non-authenticated communication (i.e., not cryptographically authenticated using component/master keys and MACs). In examples, commands directed to the second default I2C address **104** may include commands to read physical operating parameters associated with the corresponding one of print material containers **16a** to **16d** (e.g., a temperature, an ink level, and a pressure level). At the expiration of the designated time period, logic circuitry package **20** ceases responding to commands directed to the second default I2C address **104**, and resumes the secure communications session for the corresponding first default I2C address **102**.

(47) As an illustrative example, after a secure communication session has been established with the first default I2C address **102** for cyan print material cartridge **16b**, printer controller **60** may direct an “activate address” command to the first default I2C address **102** for cyan print material cartridge **16b**, wherein the “activate address” command designates a time duration. Upon receipt of the “activate address” command, logic circuitry package **20** is configured to cease responding to commands directed to the corresponding first default I2C address **102** for cyan print material cartridge **16b** for the designated time duration, and to instead to provide responses to commands directed to the second default I2C address **104** of cyan print material cartridge **16b**.

(48) In one example, the second default I2C address **104** of each print material cartridge **16a** to **16d** is a same I2C address. As such, in one example, in order to avoid potential conflict with the second default I2C address **104** of the remaining print material cartridges (**16a**, **16c**, **16d**), during the time duration, printer controller **60** directs a “change address” command to the second default I2C **104** address of cyan print material cartridge **16b**, the command including a new second I2C address **104-1**. In response to such “change address” command, logic circuitry package **20** sets the second default I2C address **104** for cyan print material cartridge **16b** to the new second I2C address **104-1** and stores the new second default I2C address **104-1** in volatile memory **120** for the duration of the

print job.

(49) Thereafter, printer controller **60** may direct commands to the new second **I2C** address **104-1** of cyan print material cartridge **16b** for the remainder of the designated time duration. In one example, printer controller **60** may direct a “read ink level” command to the new second **I2C** address **104-1** of cyan print material cartridge. In one example, logic circuitry package **20** maintains in nonvolatile memory **100** an estimated ink level value **110** for each first default **I2C** address. In one example, logic circuitry package **20** adjusts the estimated ink level for each first default **I2C** address based on factors such as the number of print sessions, and an accumulated duration of print sessions. In one example, logic circuitry package **20** reads an ink level from an ink level sensor disposed in one of the print material cartridges **16a** to **16d**, such as ink level sensor **111** in black print material cartridge **16a**, and estimates from the black ink level an ink level for each of the remaining print material cartridges **16b** to **16d** (e.g., via corresponding look-up tables). Continuing with the above example, in response to the “read ink level” command directed to the new second default **I2C** address **104-1** of cyan print material cartridge **16b**, logic circuitry package **20** provides a non-authenticated response message to printer controller **60** including the estimated ink level **110** as read from non-volatile memory **100**.

(50) In examples, printer controller **60** may monitor a status of each ink color as it is deposited onto print media **74** by printhead **72**. In one example, based on such monitoring, when printer controller **60** detects that a particular color of ink has run out, printer controller **60** provides an “out-of-ink” indication to a user, directs a message to the first default **I2C** address corresponding to the particular color to provide notification to the print material cartridge **16** that it is out of ink, and treats the print material cartridge as being out of ink. In one example, in response to the “out-of-ink notification” command, logic circuitry package **20** sets an out-of-ink flag **112** in nonvolatile memory **100** for the corresponding first default **I2C** address.

(51) Upon being powered-up for the next print job, logic circuitry package **20** assumes that the corresponding print material cartridge **16** has been replaced and resets the corresponding out-of-ink flag **112** and sets the corresponding estimated ink level value **110** to a value indicative of a “full” level, regardless of whether the corresponding print material cartridge **16** has been replaced. It is noted that since ink level value **110** and out-of-ink flag **112** are separate parameters, during a print job, ink level value **110** may indicate a non-zero ink level while the out-of-ink flag **112** indicates an out-of-ink condition, and vice-versa. Additionally, upon being powered-up for the next print job, logic circuitry package provides a new Unique ID **106** (such as from a look-up table in non-volatile memory **100**) and generates new component keys **28** and component key identifiers **108** for the “new” print material cartridge **16**.

(52) In one example, where logic circuitry package **20** is disposed within black print material cartridge **16a**, when printer controller **60** provides indication to a user and directs an “out-of-ink” message to the first default **I2C** address corresponding to black print material cartridge **16a**, logic circuitry package **60** sets the out-of-ink flag bit in non-volatile memory **100** for the corresponding first default **I2C** address **102**. Upon being powered-up for the next print job, logic circuitry package **20** assumes that black print material cartridge **16a** has been replaced and resets the corresponding out-of-ink flag and sets the corresponding estimated ink level value **110** to a value indicative of a “full” level, regardless of whether the black print material cartridge **16a** has replaced. When black print material cartridge **16a** is replaced, the ink level values **110** corresponding to each of the print material cartridges **16a** to **16d** as stored in nonvolatile memory **100** of the logic circuitry package **20** of the replacement black print material cartridge **16a** each have a value indicative of a “full” level, regardless of the actual ink level in print material cartridges **16b-16d**.

(53) In another case, continuing with the above example, during the time duration designated by the “activate address” command, printer controller **60** may direct a “read temperature” command to the new second **I2C** address **104-1** cyan print material cartridge **16b**. In one example, logic circuitry package **20** maintains in nonvolatile memory **100** an estimated temperature value **114** for each first

default I2C address **102**. In one example, the estimated temperature value **114** for each first default I2C address **102** vary from one another and are incrementally greater than an expected ambient temperature. In one example, logic circuitry package **20** reads a temperature from a temperature sensor disposed in one of the print material cartridges **16a** to **16d**, such as a temperature sensor **115** in black print material cartridge **16a**, and estimates from the measured temperature of black print material cartridge **16a** a temperature level of each of the remaining print material cartridges (e.g., via corresponding look-up tables). Continuing with the above example, in response to the “read temperature” command directed to new second I2C address **104-1** of cyan print material cartridge **16b**, logic circuitry package **20** provides a non-authenticated response message to printer controller **60** including the estimated temperature level **114** corresponding to cyan print material cartridge **16b** as read from nonvolatile memory **100**.

(54) In another case, continuing with the above example, during the time duration designated by the “activate address” command, printer controller **60** may direct a “read pressure” command to the new second default IC address **104-1** of cyan print material cartridge **16b**. In one example, logic circuitry package **20** reads a pressure level of a reservoir **90** via a pressure sensor disposed in one of the print material cartridges **16a** to **16d**, such as a pressure sensor **118** providing a pressure level of reservoir **90a** in black print material cartridge **16a**. In one example, logic circuitry package **20** estimates from the measured pressure level of black print material cartridge **16a** a pressure level of each of the remaining print material cartridges **16b-16d**, such as via corresponding look-up tables, where the estimated pressure levels corresponding to print material cartridges **16a** to **16d** are configured to vary slightly from one another and from the measured pressure level of black print material cartridge **16a**. Continuing with the above example, in response to the “read pressure” command directed to new second default I2C address **104-1** of cyan print material cartridge **16b**, logic circuitry package **20** provides a non-authenticated response message to printer controller **60** including the estimated pressure level of cyan print material cartridge **16b** as based on the measured pressure level of black print material cartridge **16a**.

(55) It is noted that because printer controller **60** controls the pressure applied to the reservoirs **90a** to **90d** of print material cartridges **16a** to **16d** via shared pump **78**, according to one example, at least one of the print material cartridges **16a** to **16d** must include a pressure sensor **118** from which logic circuitry package **20** is able to obtain a pressure level reading in order to provide estimated pressure levels for the remaining ones of the print material cartridges **16a** to **16d**. In other examples (not illustrated), in lieu of employing a pressure sensor in at least one of the print material cartridges **16a** to **16d**, logic circuitry package **20** could instead be configured to monitor pump **78**. For example, logic circuitry package **20** may employ an inductive current sensor to determine when pump **78** is active, and may employ a strain gauge to monitor a pressure on the air line **79**. Other suitable implementations may be employed to enable logic circuitry package **20** to monitor a pressure provided by pump **78**, both with and without a pressure gauge **118** disposed within one of the print material cartridges **16a** to **16d**.

(56) In other examples, printer controller **60** may direct a “slot detect” command to one of the first default I2C addresses **102** corresponding to one of the print material cartridges **16a** to **16d**, where the command instructs that the SDA line of serial data bus **80** be driven low for a time duration. Because each of the first default I2C addresses correspond to a different one of the cartridge slots **92a** to **92d**, and because a different series-connected resistance value is disposed in the SDA line between each of the cartridge slots **92a** to **92d** and printer controller **60** due to divider resistors **88a** to **88d**, printer controller **60** expects a different predetermined voltage level to be present at printer controller **60** for each of the first default I2C addresses as a result of the voltage signal of a known defined voltage level being placed on serial data bus **80** in response to the “slot detect” command. Thus, the resulting voltage at printer controller **60** from application of the voltage signal applied to the SDA line of serial data bus **80** in response to the “slot detect” command can be used by printer controller **60** to determine the cartridge slot **92a** to **92d** to which the first default I2C address

designated by the “slot detect” command corresponds. For a given one of the first default I2C addresses **102** designated by the “slot detect” command, if the resulting voltage level at printer controller **60** matches the expected predetermined voltage level, the corresponding one of the print material cartridges **16a** to **16d** is deemed to be positioned in the proper one of carriage slots **92a** to **92b**.

(57) In one example, with reference to FIG. **3**, logic circuitry package includes a voltage control circuitry **130** configured to adjust a voltage level placed on the SDA line by logic circuitry package **20** based on the one of the first default I2C addresses **102** designated by the “slot detect” command such that the resulting voltage level on the SDA line at printer controller **60** matches the expected predetermined voltage level. In one example, when logic circuitry package **20** is disposed at cartridge slot **92a**, either as part of print material cartridge **16a** or separately from print material cartridge **16a** (e.g., see FIG. **4**), voltage control circuitry **130** may comprise a resistor network, where different resistor values may be selected by control logic **30** to be connected in series with divider resistor **88a** depending on the one of the first default I2C addresses **102** designated by the “slot detect” command so that the resulting voltage at printer controller **60** matches the expected predetermined voltage level.

(58) In other examples, where logic circuitry package **20** may be disposed both at cartridge slot **92a** and at locations other than cartridge slot **92a**, voltage control circuitry **130** may comprise a controllable voltage source controlled by control logic **30** so as to provide a voltage level at the connection point of voltage control circuitry **130** to the SDA line of serial data bus **80** depending on the one of the first default I2C addresses **102** designated by the “slot detect” command so that the resulting voltage at printer controller **60** matches the expected predetermined voltage level.

(59) FIGS. **4A** and **4B** are block and schematic diagrams generally illustrating logic circuitry package including a flexible wiring substrate **140** for connecting the logic circuitry package **20** to serial data bus **80**, according to one example. In particular, FIGS. **4A** and **4B** are cross-sectional views through a portion of cartridge slot **92a** and print material cartridge **16a**.

(60) With reference to FIG. **4A**, contact **94a-1** represents an individual contact of cartridge slot contacts **94a** (see FIG. **2**) connected to the SDA line of serial data bus **80**, and contact **96a-1** represents a corresponding individual contact of cartridge contacts **96a** of print material cartridge **16a**. In one example, logic circuitry package **20** is disposed on a first surface **142** of a flexible substrate **140** along with a contact **144-1** corresponding to contact **94a-1**. A contact **146-1** corresponding to contact **96a-1** is disposed on an opposing second surface **143** of flexible substrate **140**. A conductive via **148-1** extends through flexible substrate **140** between first surface **142** and opposing second surface **143**. Logic circuitry package **20** is connected to contact **144-1** by a conductive trace **150-1** to thereby interrupt and route the SDA line of serial data bus **80** to logic circuitry package **20**. Conductive traces **148-1** and **154-1**, by way of via **148-1**, connect logic circuitry package **20** contact **146-1**, which in-turn, connect to contact **96a-1** to thereby connect logic circuitry package **20** to pressure sensor **118** within print material cartridge **16a**.

(61) FIG. **4B** generally illustrates an example logic circuitry package **20** being connected in parallel with print material cartridge **16a** to the Vdd, SCL, and GND lines of serial data bus **80**. A contact **144** on first surface **142** connects to a corresponding contact **94** of cartridge slot **92a**, and a contact on **146** on opposing second surface **143** connects to a corresponding contact **96** of print material cartridge **16a**. A via **148** extending through flexible substrate **140** from first surface **142** to second surface **143** connects contact **144** to contact **146**, and a conductive trace **150** connects logic circuitry package **20** with contact **144** such that logic circuitry package **20** and print material cartridge **16a** are connected in parallel to Vdd, SCL, and GND lines of serial data bus **80**. It is noted that such arrangement is repeated for each of the Vdd, SCL, and GND lines of serial data bus **80**.

(62) Although described primarily in terms of replaceable print material cartridges **16a** to **16d**, it is noted that logic circuitry package **20** may respond to messages directed to addresses corresponding

to any number of different types of replaceable print apparatus component other than print material cartridges.

(63) Although specific examples have been illustrated and described herein, a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present disclosure. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this disclosure be limited only by the claims and the equivalents thereof.

Claims

1. A logic circuitry package for a print apparatus comprising: a serial data bus interface to interface with a serial data bus, the logic circuit to respond to messages from the print apparatus over the serial data bus directed to a number of different addresses using different component keys for cryptographic authentication corresponding to each different address, the different component keys related to at least one master key of the print apparatus and calculated for each different address from the at least one master key and based on a unique ID associated with each of the different addresses; and control logic, in response to a session initiation request message on the serial data bus from the print apparatus to establish a secure communication session related to one of the different addresses, the control logic to cryptographically authenticate responses to each subsequent message of the secure communication session from the print apparatus using a component key corresponding to the address.
2. The logic circuitry package of claim 1, where the logic circuitry package is a single package component with a single data contact to interface with the serial data bus, configured to transmit the messages to the different addresses, and the responses to those messages, over the single data contact.
3. The logic circuitry package of claim 1, for each of the different addresses, in response to receiving a session initiation request message to establish a secure communication session with the address, the logic circuit to: generate a component session key and a component session key identifier related to a component key corresponding to the address; and provide a session response message on the serial data bus including the component session key identifier.
4. The logic circuitry package of claim 3, for each of the different addresses, after receiving the session initiation request message to establish the secure communication session, in response receiving an authenticated request message directed to the address, the authenticated request message including a first message authentication code related to a master session key associated with the master key and the component session key identifier, the logic circuit to: generate a second message authentication code related to the component session key and to the first message authentication code; and provide a cryptographically authenticated response message including response data and the second message authentication code.
5. The logic circuitry package of claim 4, when the authenticated request message comprises a read request for print data, the logic circuit to provide response data comprising print data related to print data identified by the read request.
6. The logic circuitry package of claim 4, when the authenticated request message comprises a write request including print data, the logic circuit to store the print data.
7. The logic circuitry package of claim 1, including a memory, the memory to store: the number of different addresses, each address corresponding to a different replaceable print component of the print apparatus, and operating data corresponding to each of the different addresses, the operating data including a number of component keys.
8. The logic circuitry package of claim 1, the serial data bus interface comprising an I2C serial data bus interface such that the different addresses comprise I2C addresses the different addresses including at least four different default I2C addresses, each of the four different default I2C

addresses having a corresponding further default I2C address, in response to an activate address command on the serial data bus directed to one of the four default addresses, the logic circuitry package to respond to messages directed to the further default I2C address and to suspend responding to messages directed to the one of the four default I2C addresses for a time duration specified in the activate address command.

9. The logic circuitry package of claim 8, during the time duration, in response to a change address command on the serial data bus directed to the further default I2C address, the logic circuitry package to: change the further default I2C address to a new I2C address designated by the change address command; provide non-cryptographically authenticated responses to messages from the print apparatus on the serial data bus directed to the new I2C address for a remainder of the time duration; and respond once again to the one of the different default I2C address after the time duration.

10. The logic circuitry package of claim 1, where each of the number of different addresses has a number of corresponding parameters including the unique ID representative of an ID of a corresponding replaceable print component, an ink level, and a page count.

11. A replaceable print material cartridge for a print apparatus, comprising: a print material reservoir; and a logic circuitry package including: a serial data bus interface to interface with a serial data bus, the logic circuitry package to respond to messages on the serial data bus from the print apparatus directed to a number of different addresses using different component keys corresponding to each different address, the different component keys related to at least one master key of the print apparatus and, pre-calculated for each different address from the at least one master key and based on a unique ID associated with each of the different addresses, a first one of the different addresses corresponding to the replaceable print material cartridge and each of the remaining different addresses corresponding to a different one of a number of additional replaceable print material cartridges; and control logic, in response to a session initiation request message on the serial data bus from the print apparatus to establish a secure communication session with one of the different addresses, to cryptographically authenticate responses to each subsequent message of the secure communication session from the print apparatus using a corresponding component key corresponding to the address.

12. The replaceable print material cartridge of claim 11, for each of the different addresses, in response to receiving a session initiation request message to establish a secure communication session with the address, the logic circuit to: generate a component session key and a component session key identifier related to a component key corresponding to the address; and provide a session response message on the serial data bus including the component session key identifier.

13. The replaceable print material cartridge of claim 12, for each of the different addresses, after receiving the session initiation request message to establish the secure communication session, in response receiving an authenticated request message directed to the address, the authenticated request message including a first message authentication code related to a master session key associated with the master key and the component session key identifier, the logic circuit to: generate a second message authentication code related to the component session key and to the first message authentication code; and provide a cryptographically authenticated response message including response data and the second message authentication code.

14. The replaceable print material cartridge of claim 11, each of the number of different addresses being a different first default I2C address, and each of the different first default I2C addresses having a corresponding second default I2C address, in response to an activate address command on the serial data bus directed to one of the first default I2C addresses, the logic circuitry package to respond to messages directed to the corresponding second default I2C address and to suspend responding to messages directed to the one of the first default I2C addresses for a time duration specified by the activate address command.

15. The replaceable print material cartridge of claim 14, during the time duration, in response to a

change address command on the serial data bus directed to the second default I2C address, the logic circuitry package to: change the second default I2C address to a new I2C address designated by the change address command; provide non-cryptographically authenticated responses to messages from the print apparatus on the serial data bus directed to the new I2C address for a remainder of the time duration; and respond once again to the one of the first default I2C addresses after the time duration.

16. The replaceable print material cartridge of claim 15, further including a pressure sensor to measure a pressure level of the print material reservoir, where the one of the first default I2C addresses corresponds to the print material cartridge, in response to a measure pressure command directed to the new I2C address, the control logic to obtain a measured pressure level from the pressure sensor and to provide the measured pressure level in a response message.

17. The logic circuitry package of claim 16, where the one of the first default I2C addresses corresponds to one of the number of additional replaceable print material cartridges, in response to a measure pressure level command directed to the new I2C address, the control logic to provide an estimated pressure level in a response message, the estimated pressure level based on a measured pressure level from the pressure sensor.

18. A logic circuitry package for a print apparatus, comprising: an I2C serial data bus interface to interface with an I2C serial data bus, the logic circuit to respond to messages from the print apparatus over the serial data bus directed to four first default I2C addresses using different component keys for cryptographic authentication corresponding to each of the four first default I2C addresses, the different component keys related to at least one master key of the print apparatus and calculated for each different address from the at least one master key and based on a unique ID associated with each of the different addresses, and each of the four first default I2C addresses corresponding to a different replaceable ink cartridge; and control logic, in response to a session initiation request message on the serial data bus from the print apparatus to establish a secure communication session with one of four first default I2C addresses, the control logic to cryptographically authenticate responses to each subsequent message of the secure communication session from the print apparatus using a component key corresponding to the one of the four first default I2C addresses.

19. The logic circuitry package of claim 18, for each of the different four first default I2C addresses, in response to receiving a session initiation request message to establish a secure communication session with the address, the logic circuit to: generate a component session key and a component session key identifier related to a component key corresponding to the address; and provide a session response message on the serial data bus including the component session key identifier, for each of the different four first default I2C addresses, after receiving the session initiation request message to establish the secure communication session, in response receiving an authenticated request message directed to the address, the authenticated request message including a first message authentication code related to a print session key associated with the master key and the component session key identifier, the logic circuit to: generate a second message authentication code related to the component session key and to the first message authentication code; and provide a cryptographically authenticated response message including response data and the second message authentication code.

20. The logic circuitry package of claim 18, each of the different four first default I2C addresses having a corresponding second default I2C address, in response to an activate address command on the serial data bus directed to one of the first default I2C addresses, the logic circuitry package to respond to messages directed to the corresponding second default I2C address and to suspend responding to messages directed to the one of the first default I2C addresses for a time duration specified by the activate address command, during the time duration, in response to a change address command on the serial data bus directed to the second default I2C address, the logic circuitry package to: change the second default I2C address to a new I2C address designated by the

change address command; provide non-cryptographically authenticated responses to messages from the print apparatus on the serial data bus directed to the new I2C address for a remainder of the time duration; and respond once again to the one of the first default I2C addresses after the time duration.
